

Kyle Tran

Hardware-first mechanical engineering student with hands-on experience across CAD, composite fabrication, PCB design, and embedded systems — from nationally ranked combat robots to an autonomous survey vessel built from scratch.

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SKILLS

Design	Fabrication	Electronics	Programming
Solidworks, Onshape, Fusion 360, KiCAD	Composite layup (fiberglass, carbon fiber), CNC machining, laser cutting, 3D printing	PCB design, soldering, motor/sensor integration, embedded systems (Raspberry Pi, RP2040)	Python, Java, Micropython

LEADERSHIP & TEACHING EXPERIENCE

President, Combat Robots Club, Burlingame - Taught CAD, soldering, electronics integration, material science, and 3D printing to high school students; secured sponsorships, managed club finances, and coordinated transportation to events - Led team to 16 podium finishes (4 first places, 7 second places, 5 third places) across 5 events	10/2023 – 06/2025
Subsystem Lead, Iron Panthers, Burlingame - Designed and prototyped a spindexter mechanism through CAD and R&D, managing a team of 10+ members across all grade levels to develop subsystem components - Collaborated with up to 5 mentors (professional field experts) on design iterations and technical decision-making	09/2021 – 05/2024
R&D Project Development of Ceramic Tools, Burlingame High School - Did research on various ceramic tools, developed designs via CAD, and 3D printed multiple prototypes - Created 10+ 3d printed tools for Burlingame High School's Ceramics classes and made them available online to 3d print	10/2023 – 05/2025

PROJECTS

Team Dihydrogen Monoxide, Combat Robotics - Designed and fabricated combat robots across multiple weight classes using Onshape CAD, CNC machining, laser cutting, and composite/3D printed components - Achieved top-50 national placements in 5 weight classes and 8 regional podium finishes at 2024 and 2025 national championships	03/2023 – 06/2025
Autonomous Pathfinding Robot, Computer Engineering - Designed chassis in Solidworks CAD and wrote MicroPython asyncio control system handling concurrent sensor processing, differential drive control, and real-time obstacle avoidance - Owned full hardware-software integration from mechanical assembly through firmware deployment	10/2025 – 12/2025
D.R.E.A.M.S — Autonomous Environmental Survey Vessel, Embedded Systems - Designing and fabricating a dual-hull catamaran from fiberglass and carbon fiber composite layups, owning the full mechanical design in SolidWorks from concept through build - Developing a custom KiCad PCB integrating a multi-domain power system (12S drive, 4S winch, 3S electronics) with battery monitoring, motor control, and sensor interfaces	01/2026 – Present

AWARDS

- 4th at Robot Combat League National Championship 2024 (Plastic Antweight Division), Robot Combat League
- 1st at CROS-March Madness (Antweight Divison), Combat Robotics of Sacramento
- 1st at BCRC Bell Brawls, Bellarmine Combat Robotics Club

EDUCATION

Mechanical Engineering, Northeastern University, Boston 3.87 GPA Current Coursework: Thermodynamics, Intro to Material Science, Statics, Probability and Statistics	09/2025 – Present
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